MARWAN TAREK

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OBJECTIVE

Software Engineer with hands-on experience in C#, Python, .NET, and SQL. Skilled in building scalable applications and integrating complex systems using clean architecture and modern development practices.

EXPERIENCE

Alstom — Application Engineer Integrated Control Center (ICC) July 2025 - Present *Cairo, Egypt*

- Developed and integrated software modules for SCADA and ATS subsystems using C# and scripting languages (VBS, SQL).
- Designed and optimized SQL Server and PostgreSQL schemas, writing complex LINQ queries for scalability and query performance.
- Designed and validated ICC application data for real-time railway control systems with a strong focus on quality and configuration management.
- Supported HMI (Human Machine Interface) development, including graphics and backend integration, using Windows Server environments.
- Collaborated in system testing (integration, V&V) and deployment, applying version control and traceability in line with agile processes.
- Ensured seamless integration with PLCs, RTUs, and industrial protocols (OPC-UA, Modbus) across teams.

Spark Systems — Back-end C# .net Developer Jan 2025 - June 2025 *Giza, Egypt*

- Built scalable back-end applications using ASP.NET Core MVC, C#, and Entity Framework Core, applying Clean Architecture and SOLID principles.
- Designed SQL Server and PostgreSQL schemas, optimized LINQ queries, and implemented RESTful APIs with Swagger and Postman validation.
- Collaborated in Agile Scrum teams, managing authentication with ASP.NET Identity and using Git Flow for version control.

Valeo — Embedded Software Engineer Intern July 2022 - October 2023 *Smart Village, Egypt*

- Designed and developed multithreaded C/C++ embedded software, focusing on scalability, performance, and hardware abstraction, and implemented/validated communication protocols, memory management, boot mechanisms, and low-level drivers.
- Performed system-level debugging, optimization, and functional/non-functional testing across embedded platforms, while ensuring compliance with modern C++ standards (C++11–23) and safety/coding standards.

EDUCATION

Bachelor of Computer Science, Helwan University September 2019 - July 2023 Grade: Very Good.

- Consistently led all university projects, culminating in successfully directing the team as the leader during the graduation project.

SKILLS

- C, C++, C#, Python, JavaScript, Relational & NoSQL Databases
- Embedded Systems, Backend Development, Algorithms & Data Structure, Problem solving.
- Languages: Fluent English, Native Arabic, Beginner French